

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Examination System</title>
</head>
<body>

<h1>Online Examination System</h1>

<form action="ExamServlet" method="post">

  <label>Name:</label>
  <input type="text" name="username" required>
  <br><br>

  <h3>Section A - Multiple Choice Questions</h3>

  <p>1. Which data structure follows the LIFO principle?</p>
  <input type="radio" name="q1" value="Queue" required> Queue<br>
  <input type="radio" name="q1" value="Stack"> Stack<br>
  <input type="radio" name="q1" value="Linked List"> Linked List<br>
  <input type="radio" name="q1" value="Tree"> Tree<br>

  <p>2. What is the time complexity of binary search on a sorted array?</p>
  <input type="radio" name="q2" value="O(n)" required> O(n)<br>
  <input type="radio" name="q2" value="O(log n)"> O(log n)<br>
  <input type="radio" name="q2" value="O(n^2)"> O(n2)<br>
  <input type="radio" name="q2" value="O(1)"> O(1)<br>

  <p>3. Which protocol is primarily used to securely transfer web pages?</p>
  <input type="radio" name="q3" value="HTTP" required> HTTP<br>
  <input type="radio" name="q3" value="FTP"> FTP<br>
  <input type="radio" name="q3" value="HTTPS"> HTTPS<br>
  <input type="radio" name="q3" value="SMTP"> SMTP<br>

  <p>4. Which OOP concept allows a class to acquire properties and methods from
  another class?</p>
  <input type="radio" name="q4" value="Encapsulation" required> Encapsulation<br>
  <input type="radio" name="q4" value="Inheritance"> Inheritance<br>
  <input type="radio" name="q4" value="Abstraction"> Abstraction<br>
  <input type="radio" name="q4" value="Polymorphism"> Polymorphism<br>

  <p>5. Which SQL command is used to remove a table completely from a
  database?</p>
  <input type="radio" name="q5" value="DELETE" required> DELETE<br>
  <input type="radio" name="q5" value="REMOVE"> REMOVE<br>
  <input type="radio" name="q5" value="DROP"> DROP<br>
```

<input type="radio" name="q5" value="CLEAR"> CLEAR

<h3>Section B - One-Word Answer Questions</h3>

<p>6. Which algorithm is commonly used to find the shortest path from a source vertex in a weighted graph?</p>

<input type="text" name="q6" required>

<p>7. What is the process of finding and correcting errors in a program called?</p>

<input type="text" name="q7" required>

<p>8. Which Java keyword is used to create a subclass from an existing class?</p>

<input type="text" name="q8" required>

<p>9. Which memory area stores objects created dynamically during program execution?</p>

<input type="text" name="q9" required>

<p>10. Which HTTP method is commonly used to submit form data to a server?</p>

<input type="text" name="q10" required>

<input type="submit" value="Submit Examination">

</form>

</body>

</html>

```

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;

@WebServlet("/ExamServlet")
public class ExamServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String username = request.getParameter("username");

        String q1 = request.getParameter("q1");
        String q2 = request.getParameter("q2");
        String q3 = request.getParameter("q3");
        String q4 = request.getParameter("q4");
        String q5 = request.getParameter("q5");
        String q6 = request.getParameter("q6");
        String q7 = request.getParameter("q7");
        String q8 = request.getParameter("q8");
        String q9 = request.getParameter("q9");
        String q10 = request.getParameter("q10");

        int score = 0;

        if ("Stack".equalsIgnoreCase(q1)) score++;
        if ("O(log n)".equalsIgnoreCase(q2)) score++;
        if ("HTTPS".equalsIgnoreCase(q3)) score++;
        if ("Inheritance".equalsIgnoreCase(q4)) score++;
        if ("DROP".equalsIgnoreCase(q5)) score++;
        if ("Dijkstra".equalsIgnoreCase(q6)) score++;
        if ("Debugging".equalsIgnoreCase(q7)) score++;
        if ("extends".equalsIgnoreCase(q8)) score++;
        if ("Heap".equalsIgnoreCase(q9)) score++;
        if ("POST".equalsIgnoreCase(q10)) score++;

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<head>");
        out.println("<title>Exam Result</title>");

```

```
out.println("</head>");
out.println("<body>");

out.println("<h1>Online Examination Result</h1>");

out.println("<h3>Name: " + username + "</h3>");

out.println("<h2>Your Score: " + score + " / 10</h2>");

out.println("<h3>Answers Entered</h3>");

out.println("1. " + q1 + "<br>");
out.println("2. " + q2 + "<br>");
out.println("3. " + q3 + "<br>");
out.println("4. " + q4 + "<br>");
out.println("5. " + q5 + "<br>");
out.println("6. " + q6 + "<br>");
out.println("7. " + q7 + "<br>");
out.println("8. " + q8 + "<br>");
out.println("9. " + q9 + "<br>");
out.println("10. " + q10 + "<br>");

out.println("</body>");
out.println("</html>");
}
}
```

Online Examination System

Name:

Section A - Multiple Choice Questions

1. Which data structure follows the LIFO principle?

- ☐ Queue
- ☐ Stack
- ☒ Linked List
- ☐ Tree

2. What is the time complexity of binary search on a sorted array?

- ☐ $O(n)$
- ☐ $O(\log n)$
- ☒ $O(n^2)$
- ☐ $O(1)$

3. Which protocol is primarily used to securely transfer web pages?

- ☐ HTTP
- ☐ FTP
- ☒ HTTPS
- ☐ SMTP

4. Which OOP concept allows a class to acquire properties and methods from another class?

- ☐ Encapsulation
- ☐ Inheritance
- ☒ Abstraction
- ☐ Polymorphism

5. Which SQL command is used to remove a table completely from a database?

- ☒ DELETE
- ☐ REMOVE
- ☐ DROP
- ☐ CLEAR

Section B - One-Word Answer Questions

6. Which algorithm is commonly used to find the shortest path from a source vertex in a weighted graph?

7. What is the process of finding and correcting errors in a program called?

8. Which Java keyword is used to create a subclass from an existing class?

9. Which memory area stores objects created dynamically during program execution?

10. Which HTTP method is commonly used to submit form data to a server?

Online Examination Result

Name: Alice

Your Score: 8 / 10

Answers Entered

1. Stack
2. $O(\log n)$
3. HTTPS
4. Abstraction
5. DROP
6. Dijkstra
7. Debugging
8. extends
9. f
10. post